

CHIBUIKEM OKONKWO

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PROFESSIONAL SUMMARY

Cybersecurity-aspiring Backend Developer with a focus on **Application Security (AppSec)** and **Secure Software Development Life Cycle (SSDLC)**. Proven experience building hardened backend infrastructures at Emutare (Western Australia) using Python and Node.js. Familiar with implementing Zero Trust principles, OAuth2/OpenID Connect, and cryptographic protocols. Armed with a strong foundation in Network Security (CCNA/CyberOps) and active hands-on experience in offensive security via CTF platforms.

TECHNICAL SKILLS

- **Security Engineering:** OAuth2, JWT, RBAC/ABAC, SQL Injection mitigation, XSS prevention, Data Encryption (AES, RSA), Rate Limiting.
- **Security Tools:** Wireshark, Nmap, Burp Suite, Metasploit, OWASP ZAP, Static/Dynamic Analysis (SAST/DAST).
- **Languages & Frameworks:** Python (FastAPI), JavaScript (Node.js Express), Bash Scripting, PowerShell.
- **Infrastructure & DevOps:** Linux (Ubuntu/Kali), Docker Security, Git, Cloud Security (GCP), RabbitMQ, Kafka.
- **Networking:** TCP/IP, VPN tunneling, Firewalls, DNS Security, Switching & Routing.

EXPERIENCE

Junior Security Software Engineer

Emutare, Perth, Western Australia (Remote) | Feb 2026 – Present

- **Large-Scale Microservices Architecture:** Scaled enterprise microservices architecture by 400%, architecting and deploying 7 core distributed backend services using Python (FastAPI) and relational databases.
- **Designed a Zero-Trust event-driven backbone** using asynchronous messaging, implementing a Public Key Infrastructure (PKI) layer to ensure secure inter-service communication.

- Contributed in architecting a **next-generation Identity & Access Management (IAM) platform**, integrating hardware-backed authentication (TPM), passkeys, and multi-factor security protocols.
 - **Hardened Secrets Management:** Spearheaded the migration of sensitive credentials to cloud-native key management vaults, enforcing strict security-at-rest protocols.
 - **Containerized and orchestrated** distributed backend services, optimizing service discovery and ensuring high availability across environments.
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Security Software Developer Intern

Emutare, Perth, Western Australia (Remote) | Feb 2025 – Jan 2026

- **Distributed Systems Implementation:** Contributed to the foundational architecture of an enterprise microservices platform, implementing scalable messaging patterns using **Node.js** and **NoSQL/AMQP** protocols.
 - **Cloud Infrastructure:** Provisioned secure cloud-native environments for staging and integration testing, enforcing strict network accessibility controls.
 - **Security Documentation & RBAC:** Authored technical **API documentation** and implemented initial **Role-Based Access Control (RBAC)** logic to secure service-to-service communication.
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Cybersecurity Intern

Prodigy InfoTech (Remote) | Nov 2024 – Dec 2024

- **Cryptographic Implementation:** Developed custom Python utilities for **AES-256 encryption** and secure hashing, demonstrating practical knowledge of data-at-rest protection.
 - **Traffic Analysis:** Utilized **Wireshark** to intercept and analyze PCAP files, identifying insecure protocols (Telnet, HTTP) and recommending TLS upgrades.
 - **Defensive Research:** Built a password strength analyzer using **entropy-based logic** to simulate dictionary and brute-force resistance.
 - **Malware Analysis:** Researched keylogging mechanics to develop signature-based detection patterns for endpoint defense.
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Penetration Tester (CTF Competitor) | 2023 – Present

- **Vulnerability Research:** Actively exploiting machines on Hack The Box and TryHackMe, focusing on Privilege Escalation and Web Exploitation.
- **OSINT:** Executed Open Source Intelligence gathering to map attack surfaces during simulated red-team engagements.

EDUCATION

University of Lagos • Lagos, Nigeria • B.Sc. (Ed.) Mathematics (Second Class Upper) • 2019 – 2024

University of the People • Online • B.Sc. Computer science (Cum Laude) • 2018 – 2024

- Relevant Coursework: Communications and Networking, Systems and Application Security, Advanced Networking and Data Security

CERTIFICATIONS & TRAINING

- Cisco Certified Network Associate (CCNA): Introduction to Networks – [Cisco Networking Academy](#)
- CCNA: Switching, Routing, and Wireless Essentials – [Cisco Networking Academy](#)
- CCNA: Enterprise Networking, Security, and Automation – [Cisco Networking Academy](#)
- CNSP (Certified Network Security Practitioner) – [The SecOps Group](#)
- CyberOps Associate – [Cisco Networking Academy](#)
- Cybersecurity Defence Toolbox – [Digital Regenesys](#)
- DevNet Associate – [Cisco Networking Academy](#)
- Python Essentials 1 – [Cisco Networking Academy](#)
- Africa CyberFest CTF Competition (2024) – Lagos, Nigeria.